DATA ANALYSIS PROJECT REPORT

Predicting Crime Type from Area, Time, and Victim Context in Los Angeles

Project Title	Predicting Crime Type from Area, Time, and Victim Context in Los Angeles
Student Name	Mohammad Nuruddin Mahmood
Student ID	25-62897-2
Course / Section	Programming in Python [Section D]
Instructor	Md. Tanzeem Rahat
Date of Submission	11 September 2026

1. Executive Summary

This project asks a simple question: can you tell what type of crime was reported in Los Angeles just from where it happened, when it happened, and a couple of basic details about the victim and whether a weapon was involved no case narrative, no investigation, nothing about the actual event itself? Working from a public LAPD dataset of over 134,000 incidents, I narrowed the target down to the five most frequently reported crime types, engineered a few time-based features (hour, time-of-day bucket, day of week), and compared a majority-class baseline against logistic regression, KNN, and random forest inside a leakage-safe scikit-learn pipeline. Logistic regression came out ahead not by much on macro-F1 (0.544 versus 0.109 for the baseline, and only a point or two above the other two models), but by a lot on training and prediction time, which ended up being the deciding factor. Vehicle theft is easy to call correctly from these features alone; the three theft-related categories are much harder to tell apart, which says more about what's missing from the feature set than about any one model being weak.

2. Introduction and Project Objectives

Knowing whether crime type follows any predictable pattern based on circumstances (rather than investigative detail) is a genuinely practical question it's the kind of thing that could feed into how a briefing gets written or how a report gets triaged, long before anyone looks at the specifics of a case. This project treats that question as a supervised, multiclass classification problem rather than a purely descriptive one: given area, time-of-day, day of week, hour, victim sex, and weapon involvement, predict which of the five most common crime types was reported.

Item	Your Content
Dataset name	City of Los Angeles Crime Data (2020-2024 subset)
Dataset source	LA Open Data Portal Crime Data from 2020 to Present, data.lacity.org (dataset ID 2nrs-mtv8)
Main project goal	Test whether area, time, and victim-context features carry enough signal to identify crime type well beyond a naive guess, and find the model that does this best once both accuracy and cost are considered.
Research Question 1	Can the five most common crime types be predicted from area, time-of-day, day of week, hour, victim sex, and weapon involvement — and by how much does that beat always guessing the most common type?
Research Question 2	Among logistic regression, KNN, and random forest, which model actually generalizes best to unseen data, and does the answer change once training/prediction time is taken into account?

Research Question 3 (optional)	Which crime types get confused with each other the most, and does that pattern make sense given what the features can and can't capture?

3. Dataset Description

Each row is a single reported crime incident in Los Angeles, covering when it was reported and when it actually occurred, the patrol area and reporting district, the crime type, the victim's sex where recorded, the premises type, the weapon involved where applicable, and a rough street level location. The file covers occurrences from January 2020 through December 2024, though see Section 10 for a caveat about how evenly that period is actually represented.

3.1 Basic Dataset Information

Property	Value
Number of rows	134,198 raw incident records (58,448 after restricting to the top 5 crime types and dropping rows missing a needed feature)
Number of columns	11 original columns, 17 after feature engineering
Data types present	Categorical (crime type, area, weapon, premises, victim sex), date/time (report and occurrence dates, packed HHMM time), numeric (area code, district number, derived hour)
Target or key variable(s)	CRIME, restricted to the 5 most frequently reported categories
Potential issues observed at first glance	WEAPON missing in ~81% of rows; inconsistent/missing VICT SEX codes; CRIME itself has 131 raw categories, heavily skewed toward a handful of them

3.2 Initial Observations

The first thing that jumped out just browsing the raw file was how skewed CRIME is 131 categories total, but a handful of them (vehicle theft, burglary from vehicle, shoplifting, plain theft, identity theft) make up a disproportionate share of all incidents. That's exactly why I restricted modeling to the top 5 instead of trying to classify all 131 of them; anything past the top handful just doesn't have enough rows to learn from. WEAPON being blank for 81% of rows looked alarming at first, until I noticed it lines up almost perfectly with which crime types typically don't involve a weapon in the first place so it reads more as 'not applicable' than 'missing data,' which changed how I decided to handle it in Section 4. VICT SEX also needed cleanup before it was usable: a mix of blanks, an 'X' placeholder, and inconsistent casing.

3.3 Data Source, License, and Ethics

This is published as open government data by the City of Los Angeles (LA Open Data Portal, dataset ID 2nrs-mtv8), so no special permission was needed beyond citing the original source, which I've done in Section 12. None of the features actually used in modeling are direct personal identifiers VICT SEX is the only field that touches anything demographic, and AREA NAME captures patrol-area location, not a home address or exact coordinates. One thing worth flagging honestly: a model built on patrol-area data can end up reflecting how much an area gets patrolled and reported on, not just how much crime actually happens there. I don't think that disqualifies the analysis, but it does mean the results shouldn't be read as a neutral measurement of crime that's covered properly, not glossed over, in Section 10.

4. Data Cleaning and Preparation

Three real cleaning decisions were needed before the data was usable for classification, each with a reason behind it rather than a default library setting.

Issue Found	Action Taken	Reason / Justification	Code Reference
Inconsistent VICT SEX labels (mixed case, blanks, an 'X' placeholder)	Standardized casing, then merged blanks, 'X', and	Didn't want the model treating 'unknown', 'x', and a blank cell as	Notebook, Cleaning section

	'UNKNOWN' into one UNKNOWN category	three different categories when they all mean the same thing	
WEAPON missing in ~81% of rows	Filled with the label 'WEAPON UNKNOWN' instead of dropping those rows	Dropping them would have thrown out most of the usable data, and most of the top-5 crime types are property crimes where a weapon usually isn't logged to begin with	Notebook, Cleaning section
Packed HHMM time values and text-formatted dates	Converted TIME OCC to numeric and DATE OCC to a real datetime, then derived HOUR OCC from it	You can't group or bucket a raw value like '1506' it needed to become an actual hour before it was useful for anything	Notebook, Feature Engineering section

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
HOUR OCC	TIME OCC is a packed HHMM integer (e.g. 1506); dividing by 100 and flooring gives the hour	Lets incidents be grouped by hour instead of a four-digit time code
TIME OF DAY	A small bucketing function maps HOUR OCC into Morning (5-11), Afternoon (12-16), Evening (17-20), or Night (21-4)	Turns 24 individual hours into four shifts that are easier to compare and roughly match how patrol shifts are talked about
DAY OF WEEK	Extracted from DATE OCC with .dt.day_name()	Tests whether crime type varies across the week, not just across the day
HAS WEAPON	Binary flag: 1 if a weapon was logged, 0 if WEAPON was missing/unknown	A lot easier for a model to use than the full ~50-category weapon list, and matches the 'not applicable' interpretation from Section 3.2

6. Analysis Methodology

- Pandas did essentially all of the loading, cleaning, grouping, and reshaping the class-distribution counts, the time-of-day crosstab, all of it.
- NumPy shows up directly (not just under the hood of pandas) in Section 8, where I recompute precision, recall, and F1 by hand from the confusion matrix instead of just trusting classification_report.
- Matplotlib produced every chart in this report.
- scikit-learn's Pipeline and ColumnTransformer handle the modeling itself one-hot encoding, scaling, and the three candidate classifiers — fit only on the training split so nothing from validation or test leaks into how the data gets transformed. RANDOM_STATE = 42 is fixed everywhere a seed is needed, so re-running the notebook reproduces the same numbers (see the Appendix for the full reproducibility notes).

7. Analysis and Visualizations

7.1 Research Question 1

Can the five most common crime types be predicted from area, time-of-day, day of week, hour, victim sex, and weapon involvement and by how much does that beat always guessing the most common type?

Before trusting any accuracy number, I needed to see how lopsided the target actually is.

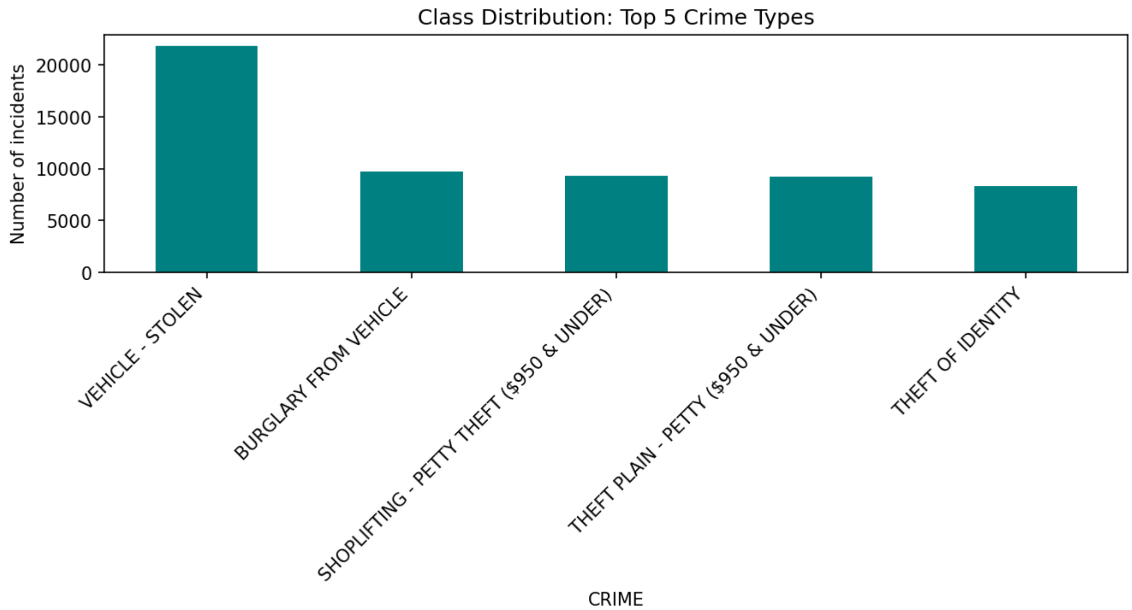

Figure 1. Class distribution of the five crime types used as the classification target.

Vehicle theft alone is 37.3% of the modeling set more than 2.6x the smallest class, theft of identity, at 14.2%. That's real, moderate imbalance, which is exactly why macro-F1 (treats all five classes equally) rather than plain accuracy is what I used to compare models.

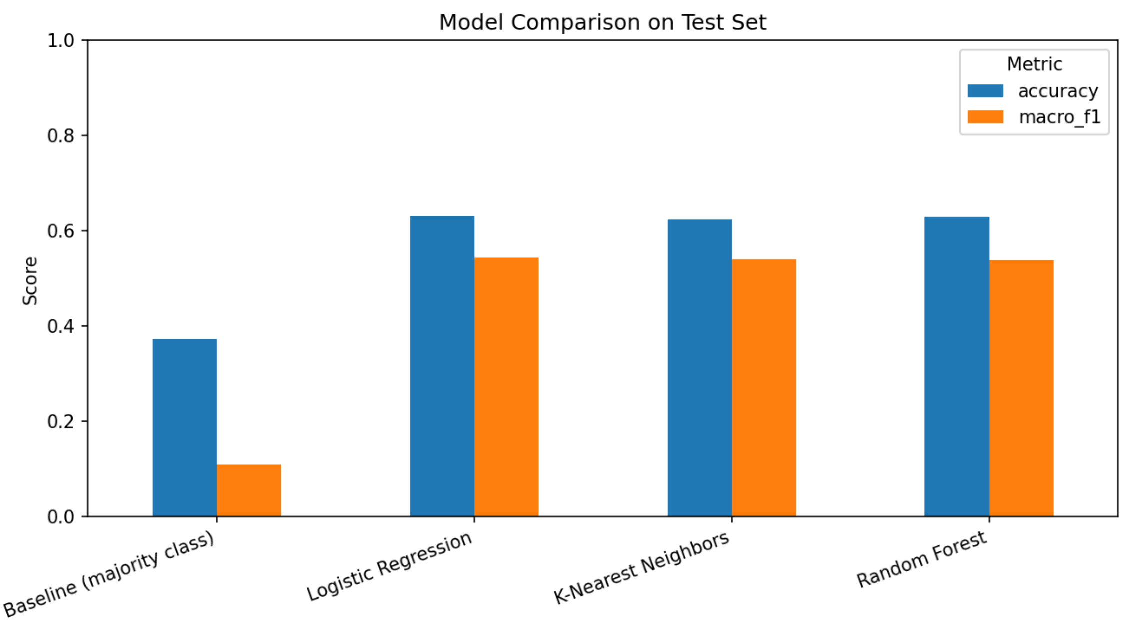

Figure 2. Test-set accuracy and macro-F1 for the baseline and all three candidate models.

The majority class baseline lands at 0.109 macro-F1 (it only ever gets the largest class right); all three real models land around 0.54 roughly five times higher. So yes, these six fairly shallow features carry a real signal about crime type, well beyond what guessing the most common category would get you.

7.2 Research Question 2

Among logistic regression, KNN, and random forest, which model actually generalizes best to unseen data, and does the answer change once training/prediction time is taken into account?

All three candidate models landed within about a point of each other on test macro-F1 (0.544, 0.541, 0.539 for logistic regression, KNN, and random forest respectively) close enough that quality alone doesn't settle the question.

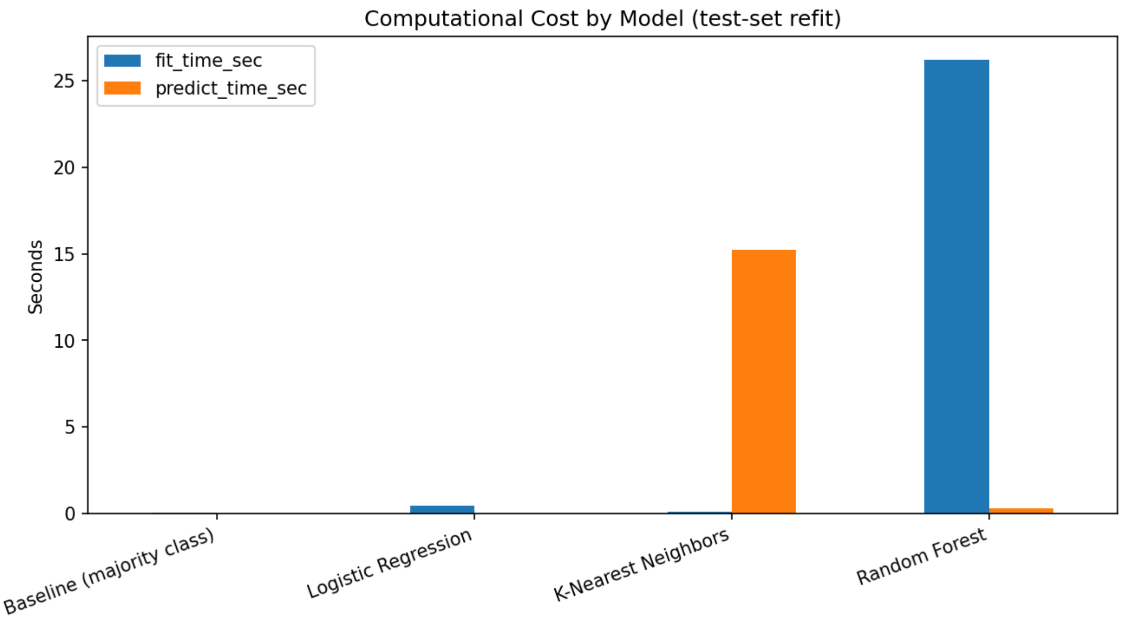

Figure 3. Fit and predict time by model, refit on train+validation and timed on the test set.

This is where it actually gets decided: logistic regression fits in well under half a second, while random forest takes 26 seconds to train and KNN takes 15 seconds just to generate predictions on the test set. Logistic regression is the model I'm defending as the final choice not because it wins on quality by very much, but because it wins on quality while being dramatically cheaper, which is a real practical difference and not a coin flip.

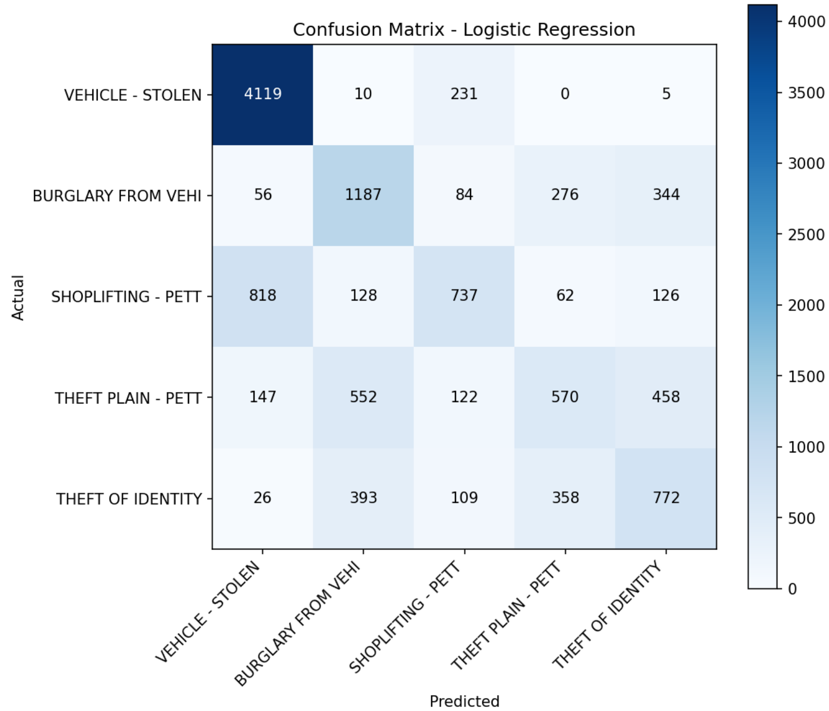

Figure 4. Confusion matrix for the selected model (logistic regression) on the test set.

Vehicle theft is by far the easiest class to get right (94% recall) it's the largest class and has a distinct night-time signature the model clearly picks up on. The three theft related categories are the weak spot: shoplifting, plain theft, and identity theft get confused with each other far more than with vehicle theft, most likely because they happen in similar areas at similar times and the current feature set genuinely can't tell them apart from six circumstantial columns alone.

7.3 Research Question 3 (optional)

Which crime types get confused with each other the most, and does that pattern make sense given what the features can and can't capture?

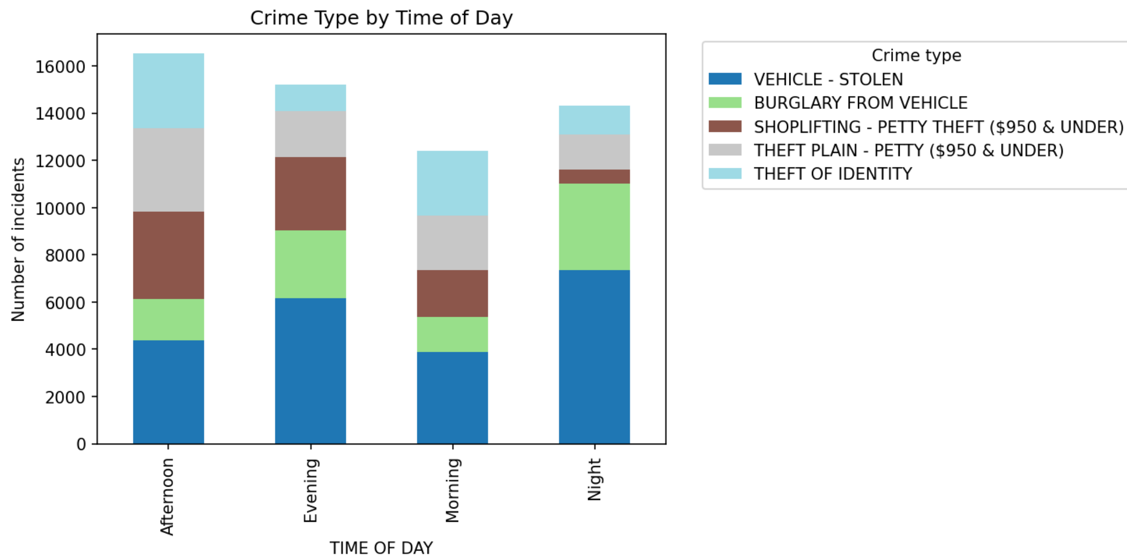

Figure 5. How the five crime types are distributed across time-of-day buckets.

Zooming out to why TIME OF DAY was worth engineering as a feature in the first place: the mix of crime types visibly shifts across the day rather than staying flat. Vehicle theft skews heavily toward Night, which tracks with cars being stolen while parked and unattended overnight, while the theft categories are relatively flatter across the day consistent with them being confused with each other in Figure 4 rather than with vehicle theft.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Manual recomputation of per class precision, recall, and F1 directly from the confusion matrix, using NumPy array operations (np.diag, elementwise division guarded against divide-by-zero) not sklearn's metric functions
Purpose	classification_report already reports these numbers, but recomputing macro-F1 by hand from just the confusion matrix is a genuinely useful way to actually understand what the metric means, rather than quoting a function name I couldn't otherwise defend in a viva
Formula or logic	For each class: TP = diagonal entry, FP = rest of that column, FN = rest of that row. precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = 2·P·R/(P+R). Macro-F1 = mean F1 across the 5 classes
Result / takeaway	Manual macro-F1 = 0.5439, matching sklearn's reported 0.5439 to four decimal places confirms the library number means exactly what I think it means

9. Key Findings

1. Logistic regression beats the majority-class baseline by roughly 5x on macro-F1 (0.544 vs. 0.109), so the six features genuinely carry signal about crime type this isn't just an artifact of one class dominating the dataset.

2. Vehicle theft is the easiest class to call correctly (94% recall): it's both the largest class and has a distinctive night-time pattern (Figure 5) that the model clearly picks up on.

3. The three theft related categories are the model's clear weak spot the confusion matrix shows shoplifting, plain theft, and identity theft getting mixed up with each other far more than with vehicle theft, most likely because they share similar areas and times that the current feature set can't distinguish.

4. Model quality alone doesn't settle which model to use here: logistic regression, KNN, and random forest are all within about a point of macro-F1, but logistic regression trains in a fraction of a second versus 26 seconds for random forest that's the deciding factor, not the accuracy gap.

5. Macro ROC-AUC for the selected model is 0.873, comfortably above the 0.5 no-skill line even where the model's single best guess is wrong, it's usually still ranking the correct class near the top, which says the underlying signal is stronger than the raw accuracy number alone suggests.

6. The manual NumPy recomputation of macro-F1 (Section 8) matched sklearn's number exactly, which is a small thing but confirms the metric is being interpreted correctly rather than just cited.

10. Limitations

- The feature set is fairly shallow area, time, day, sex, and a weapon flag can't capture premises type, exact location, or anything about the actual incident, which is probably the main reason the three theft categories get confused with each other in Section 7.2.
- Classes are imbalanced (vehicle theft is over 2.6x the size of theft of identity); macro-F1 accounts for that in how models are compared, but it doesn't change the fact that there's simply less data to learn the smaller classes from.
- This is a random stratified split, not a time-based one, so these numbers describe generalization to unseen rows from the same time range not necessarily to genuinely future incidents if reporting patterns or the crime mix shift over time.
- AREA NAME is a proxy, not a cause: patrol area frequency reflects historical patrolling and reporting intensity as well as actual crime, so the model can't separate 'more crime happens here' from 'more crime gets recorded here.' Using area based predictions operationally risks reinforcing whatever reporting bias already exists rather than describing crime itself.
- VICT SEX is included because it carries real signal in this historical dataset, not as a claim that sex predicts victimization in general using it in any operational scoring would need separate fairness review well beyond what this project does.
- Everything reported here is a correlation, not a cause. Knowing that vehicle theft peaks at night doesn't explain why it does, and none of this should be used to decide anything about a specific person or a specific patrol assignment at most it's a coarse, aggregate description of how recorded crime has historically distributed across time and area in this one dataset.

11. Conclusion

The objective here was to see whether a handful of circumstantial features area, time, day, hour, victim sex, weapon involvement carry enough signal to identify which of LA's five most common crime types occurred, without any actual case detail. They do, clearly: logistic regression reaches 0.544 macro-F1 against a 0.109 baseline, roughly five times better than guessing, and it does so faster than either alternative candidate. Vehicle theft is reliably identifiable from these features; the three theft-related categories are not well separated from each other, which looks like a feature-availability problem rather than a modeling one.

If I had more time or more permissible data, the most promising next step would be adding premises type and a coarser, privacy-conscious location feature (a grid cell rather than a street address) to see if that helps specifically with the theft vs theft confusion identified in Section 7.2. A time-based split (train on earlier years, test on later ones) would also give a more honest read on how well this generalizes to genuinely future incidents rather than just unseen rows from the same period.

12. References / Dataset Source

City of Los Angeles Crime Data from 2020 to Present. LA Open Data Portal, data.lacity.org. Dataset ID 2nrs-mtv8. https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-Present/2nrs-mtv8

pandas, NumPy, Matplotlib, and scikit learn documentation (exact versions used are listed in requirements.txt and printed at the top of the notebook).

13. Appendix

13.1 Reproducibility Notes

Item	Detail
Random seed	RANDOM_STATE = 42, set once and reused for every split and every stochastic model
Environment	Python 3.12; pandas, numpy, matplotlib, scikit-learn, jupyter, nbconvert, ipykernel pinned in requirements.txt and printed at the top of the notebook
Run instructions	See README.md restart the kernel and run all cells top to bottom, roughly 1-2 minutes total
File structure	Crime_Type_Classification.ipynb, Data/Crime_Data.csv, Figures/, requirements.txt, README.md, this report
Validation discipline	The test set was touched exactly once, after every tuning decision was finalized on the validation split

13.2 AI-Assistance Disclosure

Claude (Anthropic) was used while working on this project to help explain concepts and to review and organize parts of the code and this report so they'd be easier to understand and present clearly. All modeling decisions, parameter choices, and conclusions above are ones I can explain and defend directly.